

Inertial sensor based inverse dynamics analysis of human motions*

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Abstract— Inverse dynamics analysis, which enables the information about joint force/torque of human body from the measured motion data, is a traditional tool in biomechanical studies. Conventionally, measurements of body motion are obtained by motion capture cameras, of which the bulkiness and high cost make it hard to capture some movements that need broad range of space. As an alternative, inertial motion sensing using cheap and small microelectronic mechanical systems (MEMS) based inertial sensors has been proposed. Inertial motion sensing allows ambulatory movement and is also free of occlusion problem. It has been actively studied for recent 2 decades, but until now, most researches have focused on measuring only the kinematics of body segments.

In this research, the performance of inertial motion sensing especially aimed at inverse dynamics analysis is studied for squat motion. Kinematic data from inertial motion sensors are used to calculate the normal ground reaction force (GRF), which is evaluated based on the force plate measurement (ground truth) and additionally compared to the estimation result using optical motion sensing system. Squat motion, which can be simply modeled as a 3-segment linkage moving in sagittal plane is targeted. The normal GRF estimation result showed high correlation and low RMSE ($R=0.93$, $RMSE<0.02$ of body weight), which guarantees enough accuracy of inertial motion sensing to be used for estimating joint forces/torques through inverse dynamics analysis about squat.

I. INTRODUCTION

The information about joint force/torque of human body while conducting certain movement, is essential to analyze the cause of the movement.[1] The information of joint force/torque, which originate the movement of joints enables deeper understanding of human motions. For example, the role of each joint and the function of related muscles can be known, and the energy analysis is also available. This joint force/torque can be directly measured by invasive method [2], but the application is extremely limited. In most cases, inverse dynamics analysis is the only way to get the joint force/torque information of human motion.

The inverse dynamics analysis is based on Newtonian dynamics, as it applies equations of motion of body segments, which are obtained by assuming each body segment as one rigid link model. This analysis requires the inertial parameters and motion information of each body segment. Inertial parameters are usually estimated based on anthropometry [3], and motion information is acquired by motion sensing systems. In many cases, GRF is additionally measured by force plate to improve the accuracy of the computation.

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There are various motion sensing methods such as vision-based optical method, inertial or magnetic sensor based method. Among these, the marker-camera based motion captures system is the most widely used one. Reflective markers are located on the major joints to be tracked, and based on the vision of multiple cameras, three dimensional position information of the markers are obtained. This optical motion sensing system helps get the robust and accurate position data, and has many users and a long history. However, it has the inherent limitation of marker occlusion and bulkiness. To overcome this problem and capture wide range of space accurately, more cameras are needed, which result in another limitation of economical restrict. Also, even for the cases when many cameras are available, natural environment such as outdoor walking are limited.

With the development of MEMS inertial sensors in the 1990s, inertial sensors were much lighter, smaller and cheaper than before so that they could be applied to human motion sensing. Inertial measurement unit (IMU) is typically composed of gyroscope, accelerometer and magnetometer, giving not only the raw data from each sensor component but also the orientation data which is estimated in the internal sensor fusion algorithm.

In many studies about inertial motion sensing, the reconstruction of kinematic data was the major interest, and the goal was to get the comparable result to the conventional optical sensing method to replace it. The estimation of one joint angle[4]–[7], multi joints or whole body pose estimation including all joints[8]–[11] were conducted. However, as kinetic information is also critical for many applications in biomechanical and clinical studies, all the kinematic variables including angular velocity and acceleration are necessary. There were some studies that estimated angular velocity, angular acceleration, position or linear velocity, but they did not focus specifically on the application to inverse dynamics analysis.[9], [11], [12]

Several researchers had tried to utilize inertial sensors for inverse dynamics analysis[10], [13]–[18], or to verify the applicability.[12], [15], [16], [19], [20] Many of them did not use the appropriate evaluation criteria, because they compared the force/torque computed from inertial motion sensor data to those computed from optical motion capture system. Among these, Ohtaki has computed the GRF using inward iteration of inverse dynamics formulation, and compared the result to the force plate measurement.[16] However, the accuracy was not good enough, as the RMSE of normal GRF at most 7.6% and the RMSE of shear GRF was at most 31% of body weight.

Inverse dynamics performance should be evaluated quantitatively based on appropriate criteria. Since the kinematic data from optical motion sensing system also contain error including measurement error and signal processing error, it cannot be the best reference for validation.

Also, for the evaluation of force/torque data, the verification in force/torque domain is the most appropriate. In this research, the result of inverse dynamics analysis was verified compared to the GRF measured by force plate. As a target motion, squat motion which can be simply modeled as 2-dimensional 3-segment linkage was adopted.

II. EXPERIMENT

A. Target motion and experimental protocol

To examine the dynamics analysis performance of inertial motion sensing, squat motion which can be simply modeled as 2-dimensional 3-segment linkage was adopted, due to its accuracy and simplicity.

Total 5 subjects, 3 males and 2 females (mean $\pm$ standard deviation, age: 24.2 ± 0.4 , mass: 62.2 ± 8.9 kg, height: 1.69 ± 0.08 m) have participated in the experiments. The subjects conducted total 25 numbers of squat motions with self-selected paces, with their feet kept static on the ground. Since the upper body including head, arms and trunk is assumed as one rigid body in the 3-segment biomechanical model, the subjects were ordered not to make relative movement between those parts of the body during squat motion. As a symmetry about sagittal plane is assumed in the model, the subjects were instructed to try to keep bilateral symmetry of the movement.

C. Experimental setup

IMU sensors and markers are located on subject's body as in the Fig. 1. Three IMUs responsible for kinematic values of 3 segments of biomechanical model were affixed to shank, thigh and backside of trunk. To minimize soft skin artifact, the most rigid region of each body segment was chosen for the sensors to be located, which were also fixed firmly by Velcro, adhesive tape, elastic rubber band and neoprene band. The local x-axes of IMU sensors were aligned normal to the sagittal plane. Total 12 markers were attached on principal joint positions on the right side of the body, and were firmly fixed using double-sided adhesive tape and elastic rubber bands.[21]

Two motion sensing systems and GRF measurement system are shown in Fig. 2. For inertial motion sensing, 9-D IMU sensors (MyAHRS, Withrobot, Korea), each of which is composed of 3D accelerometer, 3D gyroscope and 3D magnetometer were used. The sensor outputs include 3D acceleration, 3D angular velocities and 3D local magnetic field which are simply the raw data, and the 3D orientation in the form of Euler angles, which is calculated through the internal sensor fusion algorithm. The bandwidth of accelerometer and gyroscope is 10Hz, and the resolution of the orientation value is 0.01 degree. IMU sensor data is acquired at 100Hz sampling frequency, using LabVIEW(National Instruments Corp., USA), through USB serial communication.

As an optical motion sensing system, V120:Trio(Optitrack, NaturalPoint, Inc., USA) which includes three cameras fixed on a rectangular frame, giving 3D position data of reflective markers with 8.33mm of latency and 0.1mm of resolution. Marker data was acquired using the software Motive 5.0. (Optitrack, NaturalPoint, Inc., USA)

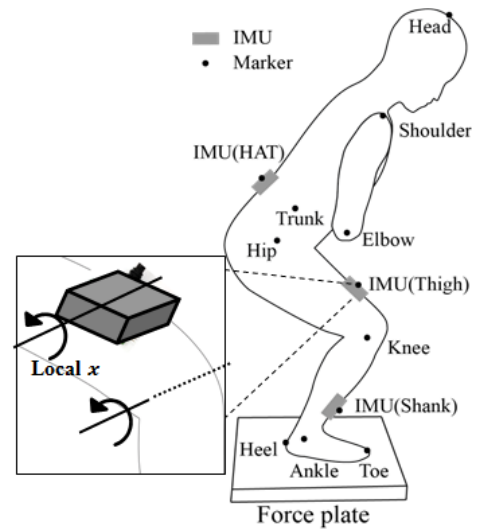


Figure 1. Marker and IMU locations for squat motion

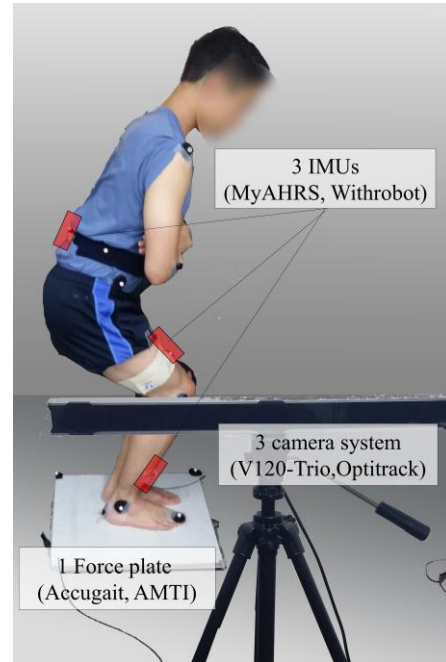


Figure 2. Experimental setup

For the reference to verify the inverse dynamics performance, GRF was also measured by force plate(AccuGait, Advanced Mechanical Technology, Inc., USA) through LabVIEW program. The analog voltage outputs of the force plate were sampled at 100Hz by DAQ board(NI PCI-6221, National Instruments Corp., USA).

III. ANALYSIS METHODS

A. Biomechanical model of squat motion

Based on the assumption of bilateral symmetry, squat motion of human body is approximated to a 3 segmented rigid linkage model moving in sagittal plane as described in the Fig.

3. [21] As a bilateral symmetry is assumed, ‘Segment 1’ represents for two shanks, ‘Segment 2’ for two thighs, and ‘Segment 3’ for the whole upper body including head, arms, and trunk, nominated as HAT segment. All the joints including the joint between the ground and ‘Segment 1’, or two joints connecting three segments are regarded as pin joints. All the inertial parameters of each segment, which include the length of the segment, center of mass, mass, moment of inertia is computed based on each subject’s height, weight, and the anthropometry table.[3]

B. Noise filtering

In order to remove measurement noise of the raw data, 5th order Butterworth low-pass filters were applied to the all raw data from three systems, IMUs, motion capture cameras, and the force plate. The major frequency components of the signals were found to be lying around 0.2 to 0.4Hz. Based on this frequency band, 5Hz was adopted as a cutoff frequency for the low-pass filtering. [22]

C. Kinematics of each segment

After the noise filtering, motion measurements from two systems were processed to produce the segment angular positions, angular velocities and angular accelerations of three segments, which are the necessary kinematic information for inverse dynamics computation. How to get these kinematic data are described in Table I.

Since the local x-axis of IMU is aligned normal to the sagittal plane, the rotation of the body segments in sagittal plane results in the rotation of the IMU about its local x-axis. The rotated angular position, represented by rotational angle output about local x-axis, stands for the segment angular position. Raw x-axis gyroscope data represents the segmental angular velocity, and the segmental angular acceleration is obtained by numerically differentiating the angular velocity value.

In case of motion capture camera, segment angular positions can be calculated based on 3D position data utilizing inverse trigonometric function as presented in the Table I. Then the angular velocities and accelerations are computed by numerical differentiation.

For the numerical calculation of derivatives, 5-point numerical differentiation method was adopted.

D. Normal GRF estimation based on downward inverse dynamics analysis

The segment angular position, velocity and acceleration data from two different motion sensing systems were put into the downward inverse dynamics computation algorithm together with the inertial model parameters, and were finally utilized to estimate the GRF. The equation of motion of the outmost segment (Segment 3) was first solved to bring reaction forces/torques between itself and the lower segment. This loop was recursively processed three times, and finally the GRF was estimated.

Originally 2-axes GRF (normal and shear GRF) and 1-axis ground reaction torque (GRT) exist for the sagittal plane motion. However, for the case of squat motion, the magnitude of shear GRF is too small to conduct error analysis, and the sagittal plane GRT could not be calculated due to the

unawareness of center of pressure (COP). Therefore, in this research, only the normal GRF was estimated and compared with the reference value, force plate measurement.

E. Error analysis

Normal GRF estimations from inertial and optical sensing systems were validated compared to the reference GRF measurement acquired from force plate. For quantitative comparison, body weight-normalized RMSE (NRMSE) and Pearson correlation coefficient (R) were computed. First, NRMSE and R for each trial were calculated, and then the mean and standard deviation of NRMSE and the mean R of 25 squat motions for each subject were calculated and organized in Table II. Mean and standard deviation of NRMSE and mean R for total 125 squat trials are also calculated and organized in Table II.

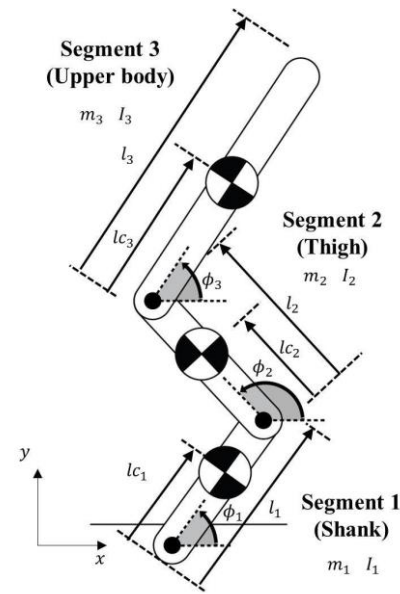
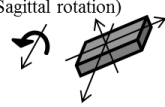
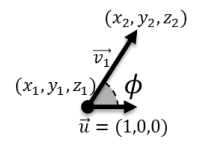


Figure 3. Biomechanical model of squat motion

TABLE I. OBTAINMENT OF SEGMENTAL KINEMATIC VALUES FROM TWO MOTION SENSING SYSTEMS

Segmental kinematics		Inertial motion sensing system	Optical motion sensing system
		Joint rotation (Sagittal rotation)  Local x	 (x_1, y_1, z_1) (x_2, y_2, z_2) $\vec{v}_1$ $\vec{u} = (1, 0, 0)$
Angular position	ϕ	Local x-axis rotational angle	Trigonometry : $\cos^{-1} \frac{\vec{v}_1 \cdot \vec{u}}{ \vec{v}_1 \vec{u} }$
Angular velocity	$\dot{\phi}$	Local x-axis gyroscope raw data	5-point numerical diff. of ϕ
Angular acceleration	$\ddot{\phi}$	5-point numerical diff. of $\dot{\phi}$	5-point numerical diff. of $\dot{\phi}$

IV. RESULTS AND DISCUSSION

Mean and standard deviation of NRMSE and mean R for total 125 squat trials are organized in Table II. Normal GRF estimation based on inertial motion sensing had a small NRMSE of 0.0180 ± 0.0024 and a high correlation coefficient of 0.926. From this result, it is inferred that motion measurements based on inertial sensors can help produce joint forces/torques for squat motion, of enough good quality.

GRF estimation from optical motion sensing showed a similar, but a slightly worse performance. NRMSE resulted in 0.0243 ± 0.0058 and the correlation coefficient was 0.864. This larger error is thought to be resulted first from the limitation of device specification. The optical motion sensing system used in this experiment was comprised of three cameras. This system is convenient to use, but may show low accuracy compared to the typical 6 or 8 camera organization that is frequently adopted in biomechanical studies.

In the Fig. 4, segmental kinematics and GRF estimation result during one randomly chosen squat trial are shown as a sample. Referring to Fig. 4.(d), the GRF estimation based on optical sensing system contains more high-frequency noises

than the estimation based on inertial sensing system. These high-frequency error peaks are obviously originated from the error peaks which can be observed in Fig. 4 (b), (c). The segmental kinematics are the bases to estimate the GRF.

In case of optical sensing, single and double differentiation of angular position was conducted to calculate angular velocities and accelerations. In comparison, for inertial motion sensing, angular velocity was directly measured and only a single differentiation was conducted to produce angular acceleration. Due to the repeated numerical differentiation, high frequency noise component of the signal is amplified.

Several previous researches tried to validate the inverse dynamics performance by comparing the joint torque estimation based on proposed system, to those based on optical motion sensing. However, the torques computed based on optical motion sensing are also the results of estimation, containing measurement error and computation error. The comparison could show the “difference” of two estimation results, but it is not reasonable to regard it as “error”. For this reason, the measurement data in force/torque domain, such as GRF would be reasonable to be used as an evaluation criterion.

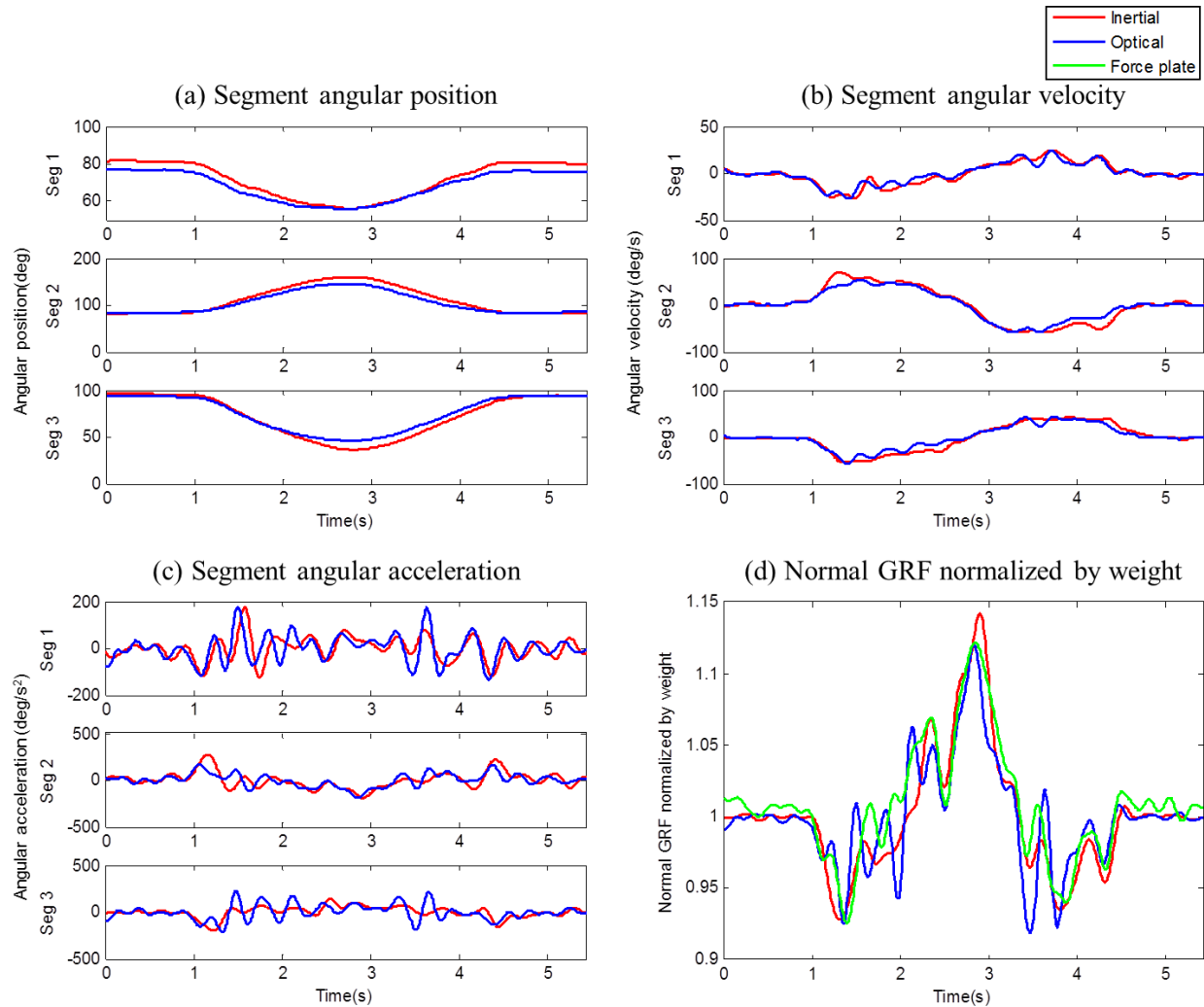


Figure 4. Kinematics for each segment: (a) Angular positions, (b) angular velocities, (c) angular accelerations and (d) normal GRF normalized by body weight during a sample trial of squat motion (Subject #5)

TABLE II. NRMSE (MEAN $\pm$ STANDARD DEVIATION) AND R VALUES(MEAN) OF NORMAL GRF USING TWO MEASUREMENT SYSTEMS – OPTICAL MOTION SENSING VS. INERTIAL MOTION SENSING

Subject No. (# of squat motions)	Normal GRF			
	Inertial motion sensing system		Optical motion sensing system	
	NRMSE (mean $\pm$ std.)	R (mean)	NRMSE (mean $\pm$ std.)	R (mean)
1 (25)	0.0158 ± 0.0022	0.936	0.0175 ± 0.0020	0.930
2 (25)	0.0196 ± 0.0027	0.921	0.0255 ± 0.0035	0.861
3 (25)	0.0157 ± 0.0022	0.952	0.0289 ± 0.0056	0.804
4 (25)	0.0193 ± 0.0016	0.895	0.0250 ± 0.0027	0.847
5 (25)	0.0195 ± 0.0033	0.926	0.0249 ± 0.0151	0.878
Total (125)	0.0180 ± 0.0024	0.926	0.0243 ± 0.0058	0.864

Using downward inverse dynamics, the accuracy of reaction torque/force estimation becomes lower for the lower segment, as the error accumulates through the iteration. In other words, the GRF, the bottommost reaction force, includes the largest error among other joint forces/torques. Therefore, the small error of the GRF estimation can guarantee the accuracy of the joint force/torque estimation using the same kinematic motion data.

V. CONCLUSION

In this research, the performance of inertial sensors for human motion sensing was studied. Kinematic data from inertial motion sensors is compared to the data from motion capture camera, and is further used to calculate normal GRF, which is evaluated based on the force plate measurements.

A squat motion, modeled as a 3-segment linkage was targeted in this research. Kinematic data measured from inertial sensing system showed similarity to the optical motion sensing system, and the normal GRF calculation based on inertial motion sensing resulted in high accuracy compared to the force plate data. This guarantees that inertial motion sensing can be applied to measure squat motion and further be used to conduct inverse dynamics analysis.

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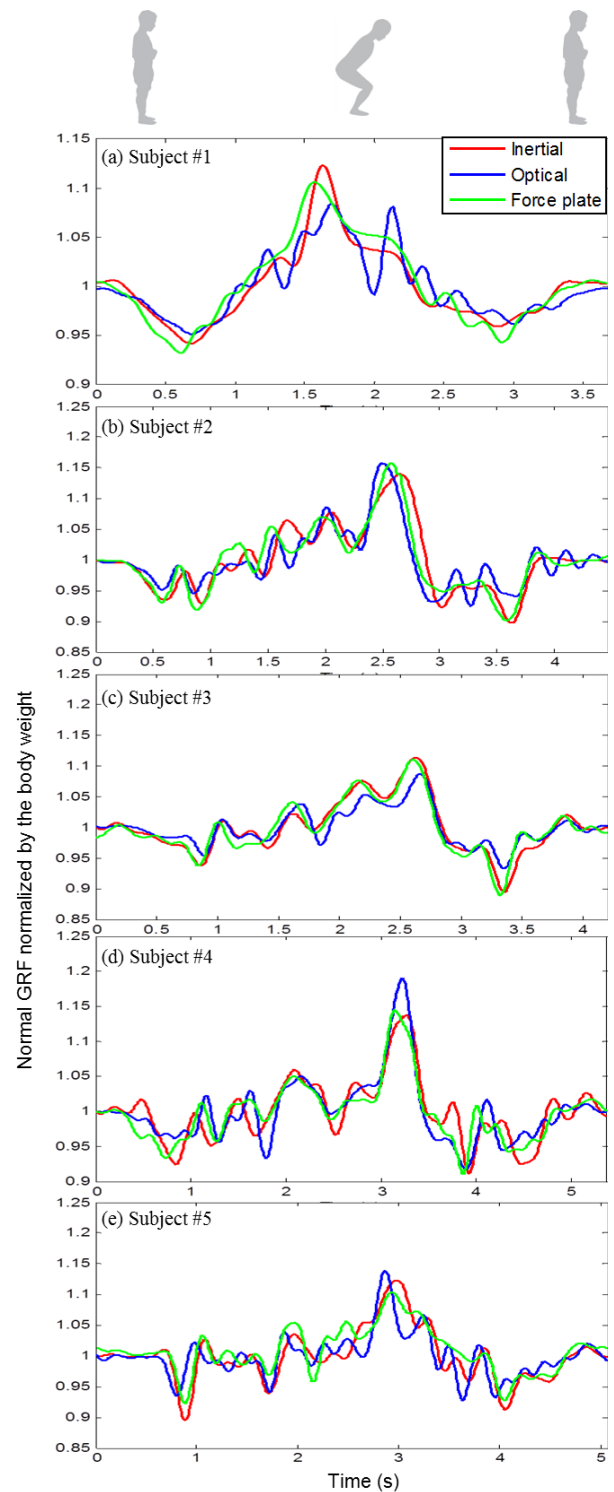


Figure 5. Normal GRF estimation from three measurement systems during one sample squat motion (where the GRF is normalized by the body weight.) (a) Subject #1, (b) Subject #2, (c) Subject #3, (d) Subject #4, and (e) Subject #5

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